Atomic GPS Base v2.0

SKU:A134-V2





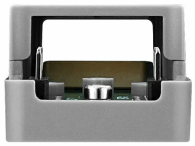


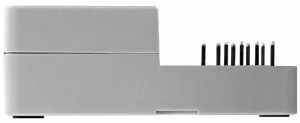




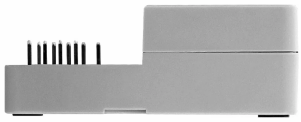














Description

Atomic GPS Base V2.0 is a GPS positioning module based on the Atom series of hosts. It integrates a high-performance AT6668 chip with a built-in MAX2659 low-noise amplifier and ceramic antenna, providing accurate and reliable satellite positioning services. It uses the NMEA0183 4.1 protocol for output and supports multi-frequency and multi-mode GNSS signal reception. It can receive signals from multiple satellite navigation systems such as GPS, BD2, BD3, GLONASS, GALILEO, and QZSS. It can be used in applications such as high-precision multi-system joint positioning and single-system independent positioning. It has strong anti-interference ability and high positioning accuracy, making it suitable for vehicle positioning and navigation, IoT positioning devices, industrial automation and robot positioning, and drone positioning.

Features

- Supports single-system positioning for multiple satellite navigation systems (GPS/QZSS/BD2/BD3/GAL/GLO) or multi-system joint positioning with any combination
- Multi-frequency and multi-system reception
- Multi-channel
- Low power consumption
- Maximum positioning update rate of 10Hz
- Development platforms:
  - Arduino IDE
  - UiFlow1
  - UiFlow2

Includes

- 1 x Atomic GPS Base v2.0

Applications

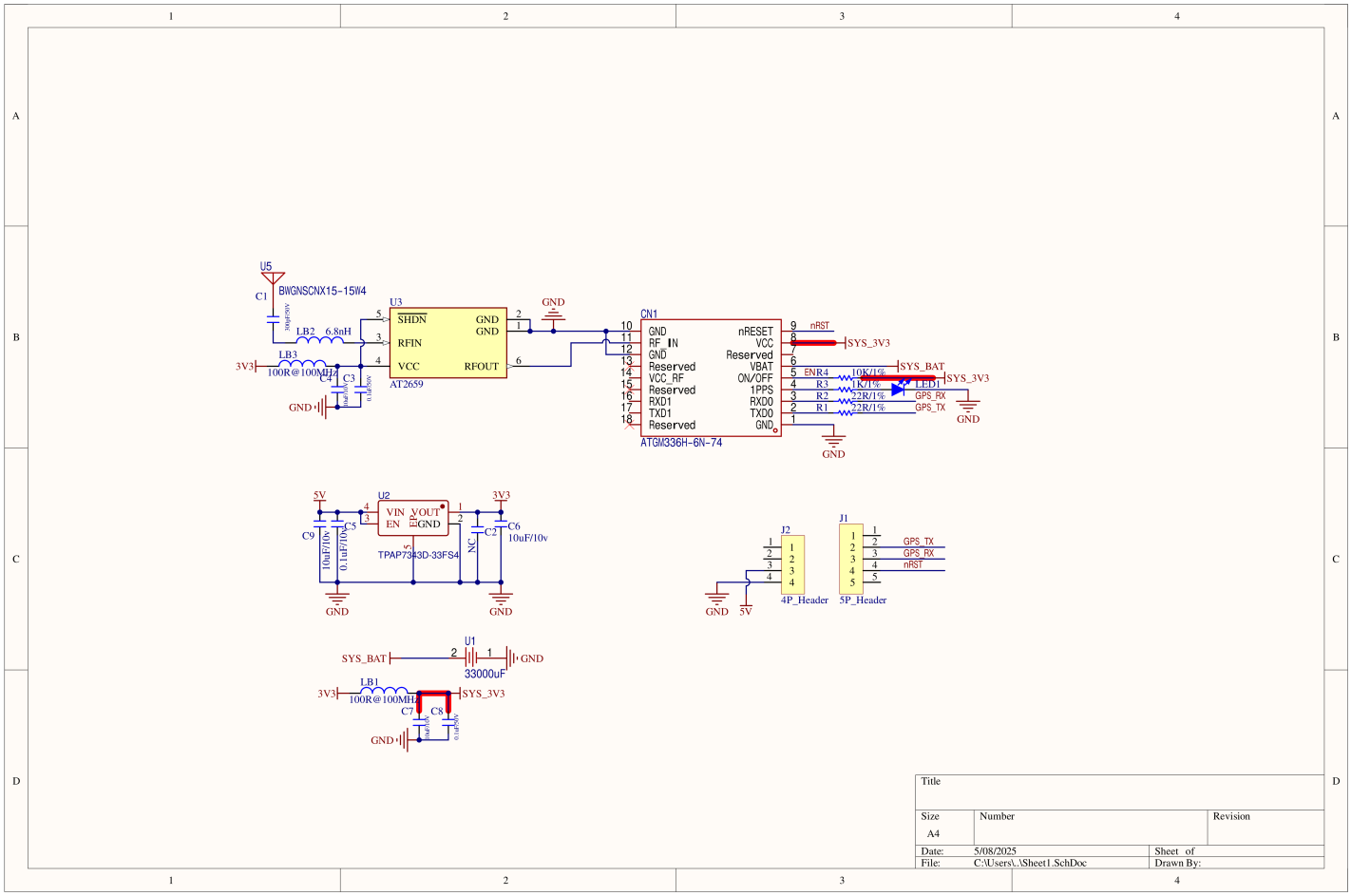
- Vehicle positioning and navigation
- Internet of Things (IoT) positioning devices
- Outdoor positioning devices
- Drone positioning

Specifications

Specification	Details
SoC	ATGM336H-6N-74 module @AT6668
Satellite Frequencies	BDS: B1, B1C GPS/QZSS: L1 GALILEO: E1 GLONASS: L1
Positioning Accuracy	<1.5m (CEP50)
Channels	50
Output Frequency	1Hz (max 10Hz)
Startup Time	Cold start: 23s; Hot start: 1s; Reacquisition: 1s
Acquisition Time	1s
Baud Rate	115200bps@8N1
Data Format	NMEA0183
Sensitivity	Tracking: -167dBm; Reacquisition: -160dBm; Cold start: -148dBm; Hot start: -156dBm
Power Consumption	Standby current: DC 5.01V@4.95mA; Operating current: DC 5.02V@35.07mA
Product Size	48.0 x 24.0 x 17.5 mm
Product Weight	13.2g
Package Size	138.0 x 93.0 x 19.0mm
Gross Weight	15.5g

Schematics

- Atomic GPS Base v2.0 Schematics PDF

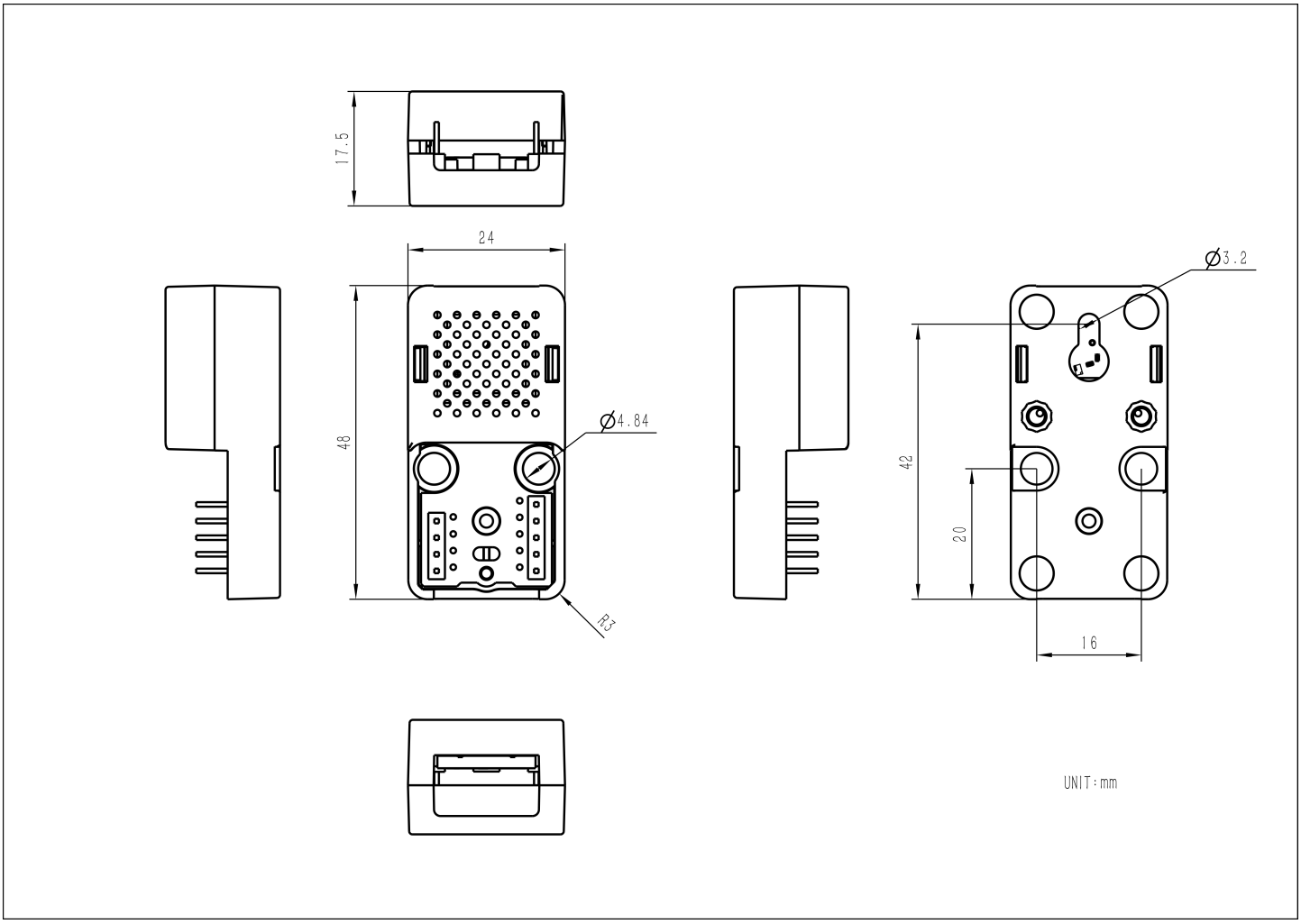


PinMap

PIN	LEFT	RIGHT	PIN
		1	NC
NC	2	3	UART_TX
NC	4	5	UART_RX
5V	6	7	RST
GND	8	9	NC

Model Size

Atomic GPS Base v2.0 Model Dimensions PDF



Datasheets

- ATGM336H-6N
- AT2659
- TPAP7343D
- CASIC Multi-Mode Satellite Navigation Receiver Protocol Specification

Softwares

Quick Start

- Atomic GPS Base v2.0 Arduino Quick Start

Arduino

- Atomic GPS Base v2.0 Arduino Library

UiFlow2

- Atomic GPS Base v2.0 UiFlow2 Docs

Video

Atomic GPS Base v2.0 product introduction and case demonstration

A134-V2-Atomic-GPS-Base-v2.0_video.mp4

Product Comparison

Product Comparison	 Atomic GPS Base v2.0	 Atomic GPS Base
Positioning Accuracy	<1.5m	2m
microSD Card Slot	No	Yes
Module Model	AT6668	M8030-KT
Supported Satellite Systems	GPS / QZSS / BD2 / BD3 / GAL / GLO	BDS / GPS / GLO / GAL / SBAS / QZSS